
WHAT IT DOES

Keeps the tent at the target climate for the selected grow profile. Every 30 s it reads temperature, humidity and CO2, scores how far each is off ideal (0-100, 50 = on target), and commands all actuators at once from that score.

Sensor and event data is written to the SD card; if the card is missing it falls back to internal flash, then to RAM, and migrates the backlog once the card returns.

BUTTON

Tap	next screen
Hold 3 s	screen action
Idle 30 s	back to screen 1
Idle 120 s	display sleeps
Any press	wakes display

OLED SCREENS (tap to cycle)

TEMPERATURE	now/hi/lo/avg, 1 h. Hold = clear
HUMIDITY	now/hi/lo/avg, 1 h. Hold = clear
SERVICE	days since service. Hold = reset
SD CARD	mount state + MB. Hold = remount
ALERTS	active ERR/WRN keys, or All OK
SYSTEM	clock, firmware, uptime, RAM
RELAYS	mains channels + PWM fan duty
REGULATION	band, deviations, commands
CO2	ppm + vent override
SOIL	moisture % (plant mode only)
DEBUG	Hold = open test menu

REGULATION SCREEN

Line 1 Band 2 ok - severity band, and mode: ok / EMERG (emergency vector) / LATCHED (held in safe state, needs a power cycle or a fixed climate to clear).

T H C deviation temp / humidity / CO2
0 = far low, 50 = ideal, 100 = far high

Lines 3-5 are commanded output, 0-100 %:

He	heater
Fo	heater follower fan
Cl	cooler
Hu	humidifier
Ex	exhaust fan
Ci	circulation fans
Gl	grow light
b	day/night blend, 0 = night, 1 = day

MAINS RELAYS (REL_CON1)

Cool GP18	cooler / AC
Humi GP19	humidifier
Lite GP20	grow light (+ 0-10 V dimmer)
Spar GP21	spare, unassigned
Screen shows ON / OFF, or -- when the channel has no controller.	

PWM FANS (PCA9685, 12 V)

ch0	circulation, centre
ch1	circulation, walls
ch2	heater follower
ch3	case fan, always on 60 %
ch4	exhaust

STATUS LEDS

Activity	blinks on each read/write
SD	off = mounted, solid = no card, blink = card present but unreadable
Warning	solid = degraded, still running
Error	solid = fault, needs attention
Service	blinks when service is due
Pico LED	heartbeat, firmware is alive

SENSORS

SHT31-D	temperature + humidity, I2C
Senseair S8	CO2 ppm, UART
DS3231	real-time clock, I2C
Soil	analog, plant mode only

ALERT KEYS (ALERTS screen)

ERR sd_required	SD needed but unavailable
ERR th_dead	climate sensor not answering
ERR mem_error	RAM critically low
ERR logged_error	see system.log on the SD card
WRN th_intermit	sensor dropping reads
WRN fallback_act	logging to flash, not SD
WRN buffer_backlog	writes queueing up
WRN mem_warn	RAM above warn level
WRN rtc_invalid	clock time implausible
WRN co2_stale	CO2 reading too old; timed venting
WRN co2_unreach	CO2 sensor silent; check its plug
WRN sht31_unreach	climate sensor silent; check its plug

FIRST AID

SD LED on	reseat the card, then SD screen, hold
LATCHED	fix climate, then power cycle
Blank screen	tap the button; it sleeps at 120 s
Nothing runs	check 12 V input and the fuse